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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CL/EC/ME) Sem - I Examination, 2009-10
PY - 101 Engineering Physics

Date: 16.12.09, Wednesday Time: 02 p.m to 05 p.m Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

- Q. 1**
- | | | |
|---------|---|---|
| (a) | How does laser differ from ordinary light? | 1 |
| (b) | A classroom has dimensions $20 \times 15 \times 5 \text{ m}^3$. The reverberation time is 3.5 Sec. Calculate the total absorption of its surface and average absorption coefficient. | 2 |
| (c) (i) | What is the general classification of errors? Give an example of each. How do we eliminate the errors? | 3 |
| (ii) | The mean diameter of a thin brass rod is to be measured by vernier callipers. Why is a set of 100 measurements of the diameter expected to yield a more reliable estimate than a set of 5 measurements? | 1 |

- Q. 2**
- | | | |
|---------|--|---|
| (a) | Mention four important conditions to be satisfied for good acoustics. | 4 |
| (b) | Discuss the properties of ultrasonic waves? How are they detected? | 5 |
| (c) (i) | Define reverberation period for a hall. Write an expression for it. How can one minimize it? | 2 |
| (ii) | Define absorption coefficient of a material and derive an expression for its determination. | 3 |

OR

- Q. 2**
- | | | |
|---------|---|---|
| (a) (i) | Distinguish between musical sound & noise. | 2 |
| (ii) | Explain the terms: (i) threshold of feeling and (ii) dead effect | 2 |
| (b) | Explain any five applications of ultrasound in brief. | 5 |
| (c) | Which method is suitable for the production of ultrasonic waves of frequency of the order of 10^3 Hz ? Describe it with a suitable diagram. | 5 |

- Q.3**
- | | | |
|---------|--|---|
| (a) (i) | Explain the reason for the monochromatic nature of laser beam. | 1 |
| (ii) | Name the components of laser system. Briefly explain each of them. | 3 |
| (b) | How does stimulated emission take place with exchange of energy between nitrogen and CO_2 atoms? Explain the operation with the help of a neat labeled diagram. | 5 |
| (c) (i) | How does holography differ from ordinary photography? | 2 |
| (ii) | Explain how hologram is prepared and reviewed. | 3 |

OR

- Q.3**
- | | | |
|---------|--|---|
| (a) | What is an optical fiber? Explain the principle of an optical fiber as a waveguide for light. Hence give the advantages of fiber optic cable. | 4 |
| (b) | Derive the expression for the numerical aperture of step index fiber. Show that it does not depend on the physical dimensions of the fiber. | 5 |
| (c) (i) | Define the relative refractive index difference of an optical fiber. Show how it is related to numerical aperture. | 2 |
| (ii) | The core of a glass fiber has refractive index 1.5. If the relative refractive index is 0.005, find refractive index of cladding, critical angle, acceptance angle and numerical aperture. | 3 |

SECTION - II

- Q.4** (a) What is a SQUID? Write two of its applications. 2
 (b) Define coordination number, atomic radii and atomic packing factor 3
 (c) What are nano materials? List any four of them. 2

- Q.5** (a) Explain any four characteristic properties of objects in the nanoscale with suitable examples. 4
 (b) (i) What do you mean by a primitive cell? 1
 (ii) Consider a body centered cubic (BCC) lattice of identical atoms having radius R . Compute the number of atoms per unit cell, coordination number and atomic packing factor. 4
 (c) (i) Derive the relation between lattice constant and density for cubic crystal system. 2
 (ii) Calculate the density of Cu which has FCC structure. Its atomic radius is $1.278 \times 10^{-10} \text{ m}$ and atomic weight is 63.54. (Avogadro's number = $6.023 \times 10^{26} / \text{Kg mol}$). 3

OR

- Q.5** (a) Write four applications of nanomaterials in the field of engineering and industry? 4
 (b) (i) Deduce the relation between inter planar distance 'd' and the Miller indices of the planes for cubic crystal. 3
 (ii) Calculate the interplanar spacing for (3, 2, 1) plane in a simple cubic lattice where lattice constant is $4.2 \text{ \AA}$. 2
 (c) (i) What are Miller indices? 1
 (ii) Sketch the following Miller index planes: (1 2 0), (2 0 0), (2 0 2) and ($\bar{1}$ 1 0). 4
- Q.6** (a) What is Hall effect? Obtain an expression for Hall coefficient for an extrinsic semiconductor. Show that the result of Hall effect may be used to deduce carrier concentration. 4
 (b) Describe the construction and operation of the Coolidge tube for the production of X-rays. How are the intensity and hardness of X-rays controlled? 5
 (c) (i) What are superconductors? Give examples. 1
 (ii) How superconductors are advantageous compared to normal conductors? 1
 (iii) What is the significance of critical temperature, critical magnetic field and critical current density for super conductors? 3

OR

- Q.6** (a) (i) Give two engineering applications of superconductors in brief. 1
 (ii) What is a Josephson junction? Explain a.c. Josephson effect and d.c. Josephson effect. 3
 (b) (i) How does Hall effect show whether holes or electrons predominate in a semiconductor? 1
 (ii) The Hall coefficient of a specimen of doped silicon is found to be $3.66 \times 10^{-4} \text{ m}^3 \text{C}^{-1}$. The resistivity of the specimen is $8.93 \times 10^{-3} \Omega \text{ m}$. Find the mobility and density of the charge carrier. 4
 (c) (i) An x-ray tube operates at 30kV. Calculate the shortest wavelength of X-rays produced. 2
 (ii) The spacing between successive planes in a crystal is $2.82 \text{ \AA}$. X-ray incident upon the surfaces of the crystal is found to give rise to the first order Bragg reflection at a grazing angle at $8^\circ 35'$. Calculate the wavelength of the X-ray and find the angle at which the second order Bragg reflection would occur. 3